

DAX RAJANI

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WORK EXPERIENCE

Glide Technology Pvt. Ltd. | Ahmedabad, India

Jul 2023 – Aug 2024

Associate Engineer, Embedded Firmware

- Designed and implemented a complete MiWi proprietary protocol connection-management module in embedded C on ARM Cortex-M under FreeRTOS, handling channel scanning, hub registration, device identifier exchange, and broadcast-mode fallback to sustain uptime across a **4,000+ node** real-time production deployment.
- Reduced wearable device power consumption by **30%** by migrating the BLE stack from Bluetooth Classic to BLE and restructuring FreeRTOS task scheduling to minimize active-state time during continuous telemetry workloads.
- Improved WiRa indoor positioning accuracy by **25%** by implementing RSSI-based distance calculation with data-frame validation filters that discarded non-compliant BLE advertising packets at the driver level in real time.
- Designed embedded C communication protocols including BLE data parsing, CRC validation, and interrupt-driven retry logic between host and slave ARM Cortex-M controllers over UART and SPI, increasing packet transfer reliability by **20%**.
- Built OTA and recovery workflows using dual-image update patterns, rollback-safe boot behavior, and staged firmware delivery; reduced host command-dispatch overhead by **15%** by developing transport-agnostic USB/UART control firmware with interrupt-driven execution and optimized state-machine transitions.

Glide Technology Pvt. Ltd. | Ahmedabad, India

Dec 2022 – Jun 2023

Software Engineering Intern, Embedded Systems

- Resolved a critical memory exhaustion defect in production BLE firmware within **2 weeks** by tracing heap allocations across a multithreaded FreeRTOS state machine, restoring stable operation to **4,000+ deployed sensors** resetting every 5-10 seconds.
- Prototyped a WiRa indoor positioning system on the DA14695 BLE SoC in embedded C with Yocto Linux, configuring BLE advertising, RSSI-based distance calculation, and data-frame validation logic for filtering malformed sensor packets before driver-level processing.
- Built practical foundation in bare-metal C, FreeRTOS, memory management, interrupt handling, Yocto Linux, I2C, SPI, UART, cross-compiler toolchains, and hardware debugging with JTAG, oscilloscope, and logic analyzer.

PROJECTS

Secure A/B OTA Bootloader | C, MCUboot, Zephyr RTOS, BLE SMP, ECDSA-P256, nRF52840, GitHub Actions2025

- Implemented production-grade OTA firmware update system for nRF52840 using MCUboot with swap-scratch A/B slots and ECDSA-P256 image signing; unsigned images cryptographically rejected at boot by embedded public key.
- Delivered firmware wirelessly via BLE SMP using nRF Connect mobile; demonstrated automatic rollback with **3-boot failure** then automatic revert via CRASH_DEMO build flag; GitHub Actions CI verifies signed binaries on every push.

Zephyr BLE Sensor Node | C, Zephyr RTOS, BLE 5, GATT, LE Secure Connections, NVS, nRF52840

2025

- Built BLE 5 peripheral firmware on nRF52840 DK using Zephyr RTOS with a custom **128-bit GATT service** exposing temperature, humidity, and configurable sample rate over an encrypted, authenticated LE Secure Connections link.
- Implemented per-characteristic CCCD subscription tracking, independent notification control, and NVS-backed settings persistence; deferred advertising restart on disconnect via Zephyr work queue to avoid BT stack re-entrancy.

E-Commerce Analytics Pipeline | Apache Spark, GCP Dataproc, Python, PySpark

Winter 2026

- Designed a distributed analytics pipeline on GCP Dataproc using Apache Spark, implementing sessionisation, funnel analysis, last-touch attribution, and anomaly detection on a **42M row dataset** across a 9-configuration cloud scaling study (3 cluster sizes x 3 data sizes).

EDUCATION

Concordia University | Montreal, QC

2024 – 2026 (Expected)

Master of Engineering, Electrical and Computer Engineering

Relevant Coursework: Applied Machine Learning, Computer Neural Networks, Biological and Medical Image Processing, Distributed Systems, Embedded Systems

Ganpat University | Gujarat, India

2019 – 2023

Bachelor of Technology, Computer Engineering

ACHIEVEMENTS

- Awarded **Emerging Star of the Year 2023-24** at Glide Technology for technical contributions across multiple embedded firmware production deployments.
- Earned **3 provincial and district gold medals** in competitive Powerlifting (2020-2023), demonstrating sustained high-performance discipline and resilience.
- Recognized as Volunteer of the Year 2023-24 at Glide Technology for organising corporate events and managing digital content production.

KEY SKILLS

Embedded Firmware: C, Embedded C, C++, Python, Bare-Metal, State Machine Design, Memory Management, Interrupt Handling, Multithreading, Modular Architecture

Microcontrollers and Hardware: ARM Cortex-M3/M4, nRF52840, Dialog DA14695, NXP MCUs, JTAG, J-Link, Logic Analyzer, Oscilloscope, Memory-Mapped I/O

RTOS and OS: FreeRTOS, Zephyr RTOS, Yocto Linux, Embedded Linux, Cross-Compiler Toolchains, Task Scheduling, Power Optimisation

Protocols: BLE 5.x, MiWi Proprietary, UART, I2C, SPI, USB, HCI, CRC Validation, Driver Development

Bootloaders and OTA: MCUboot, A/B Swap, ECDSA-P256 Image Signing, imgtool, DFU, Secure Boot

AI and ML: PyTorch, scikit-learn, XGBoost, NumPy, Matplotlib, Ensemble Methods, Applied ML, Neural Networks

Tools and SDLC: Git, GitHub Actions, nRF Connect SDK, west, GDB, GCP Dataproc, Apache Spark, Agile Scrum, Code Reviews, Technical Documentation